🌱 Week 1 Git & GitHub Tutorial: Version Control for Developers

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🔷 1. What is Git?

- **Git** is a **distributed version control system**.
- It tracks changes to code and allows multiple developers to collaborate.
- You can **go back in time**, compare versions, create branches, and merge changes.

🔷 2. What is GitHub?

- **GitHub** is a **cloud-based platform** that hosts Git repositories.
- It enables collaboration using features like:
    - **Pull requests**
    - **Issues**
    - **Branch protection**
    - **Project tracking**

> Think of Git as the engine, and GitHub as the remote dashboard to work with it.

🔷 3. Initial Git Setup (One-time)

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

This sets up your Git identity.

🔷 4. Creating a New Repository

✅ Step 1: Initialize a Git repo

```
git init
```

> This turns the current directory into a Git-tracked project.

🔷 5. Tracking Changes

✅ Step 2: Check status

```
git status
```

> Shows what's new, modified, or staged.

✅ Step 3: Add changes to the staging area

```
git add .
```

> Stages **all** changes for the next commit.

You can also stage a single file:

```
git add filename.java
```

🔷 6. Making a Commit

✅ Step 4: Commit your changes

```
git commit -m "Initial commit"
```

> Takes a snapshot of the staged files with a message.

📝 Tip: Always use clear and concise commit messages!

🔷 7. Viewing History

```
git log
```

> Shows a history of commits (author, date, message, hash).

🔷 8. Working with Remote GitHub Repo

✅ Step 1: Create a repo on GitHub

Go to https://github.com and click **New Repository**

✅ Step 2: Connect local repo to GitHub

```
git remote add origin https://github.com/your-username/your-repo.git
```

✅ Step 3: Push your local code to GitHub

```
git push -u origin main
```

> The `-u` flag sets the upstream so future pushes can just be `git push`.

🔷 9. Cloning a Repo

If you’re joining a team or downloading code:

```
git clone https://github.com/username/project.git
```

> Copies the full repo to your local machine.

🔷 10. Creating and Switching Branches

```
git checkout -b feature-xyz
```

> Creates and switches to a new branch called `feature-xyz`.

🔷 11. Pushing a Branch

```
git push origin feature-xyz
```

> Uploads your new branch to GitHub.

🔷 12. Submitting a Pull Request (on GitHub UI)

1. Push your branch to GitHub
2. Go to GitHub and click “Compare & Pull Request”
3. Add a title & description, then submit

🔷 13. Summary of Commands

Command	Purpose
`git init`	Initialize a Git repo
`git status`	Check file changes

Command	Purpose
`git add .`	Stage all changes
`git commit -m "msg"`	Save changes with message
`git log`	View commit history
`git remote add origin URL`	Link local repo to GitHub
`git push -u origin main`	Push code to GitHub
`git clone URL`	Download a GitHub repo
`git checkout -b branchname`	Create and switch to new branch
`git push origin branchname`	Push branch to GitHub

🧪 Lab Practice

1. Initialize a repo
2. Write and commit a simple Java program
3. Create a GitHub repo and push your code
4. Create a feature branch and push it
5. Submit a pull request